

Time Series Forecasting for Portfolio Management

Interim Submission | GMF Investments | 2026-07-05

TSLA • BND • SPY • ARIMA • Stationarity • Risk Metrics

From Raw Prices to a Stationarity-Aware Forecasting Baseline

This interim submission covers Task 1 (data extraction, cleaning, exploratory analysis, stationarity testing, and risk metrics) and the first model of Task 2 (a chronologically validated ARIMA baseline for TSLA) for GMF Investments' portfolio forecasting initiative. All figures and numbers below are taken directly from the executed project notebooks.

2,888 Trading days per asset	d=1 Differencing order confirmed by ADF	ARIMA(0,1,0) Selected order (auto_arima)	17.11% Test-period MAPE
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1. Data Extraction and Cleaning

Daily OHLCV data for **TSLA**, **BND**, and **SPY** were fetched via `yfinance` for 2015-01-01 through 2026-06-30 (`src/data_loader.py`). Yahoo Finance returned 2,888 trading days per ticker, from 2015-01-02 through 2026-06-29.

- **Dtype/schema check:** all columns loaded with correct numeric/datetime types; no null values were present in any of the returned trading-day rows.
- **Calendar alignment:** prices were reindexed onto a common business-day calendar across all three tickers so they can be merged and compared on the same date axis.
- **Gap handling:** any gaps introduced by reindexing (e.g. asset-specific holidays) were **forward-filled** rather than interpolated — a price is genuinely unchanged until the next trade occurs, so forward-fill reflects reality, whereas interpolation would fabricate price movement that never happened.
- **Feature scaling:** daily percentage returns were computed from adjusted close prices as the basis for all volatility and risk calculations; price levels were separately min-max normalized purely for plotting three series with very different scales (TSLA ~\$400 vs. BND ~\$70) on one chart.

2. Exploratory Data Analysis

2.1 Price trends

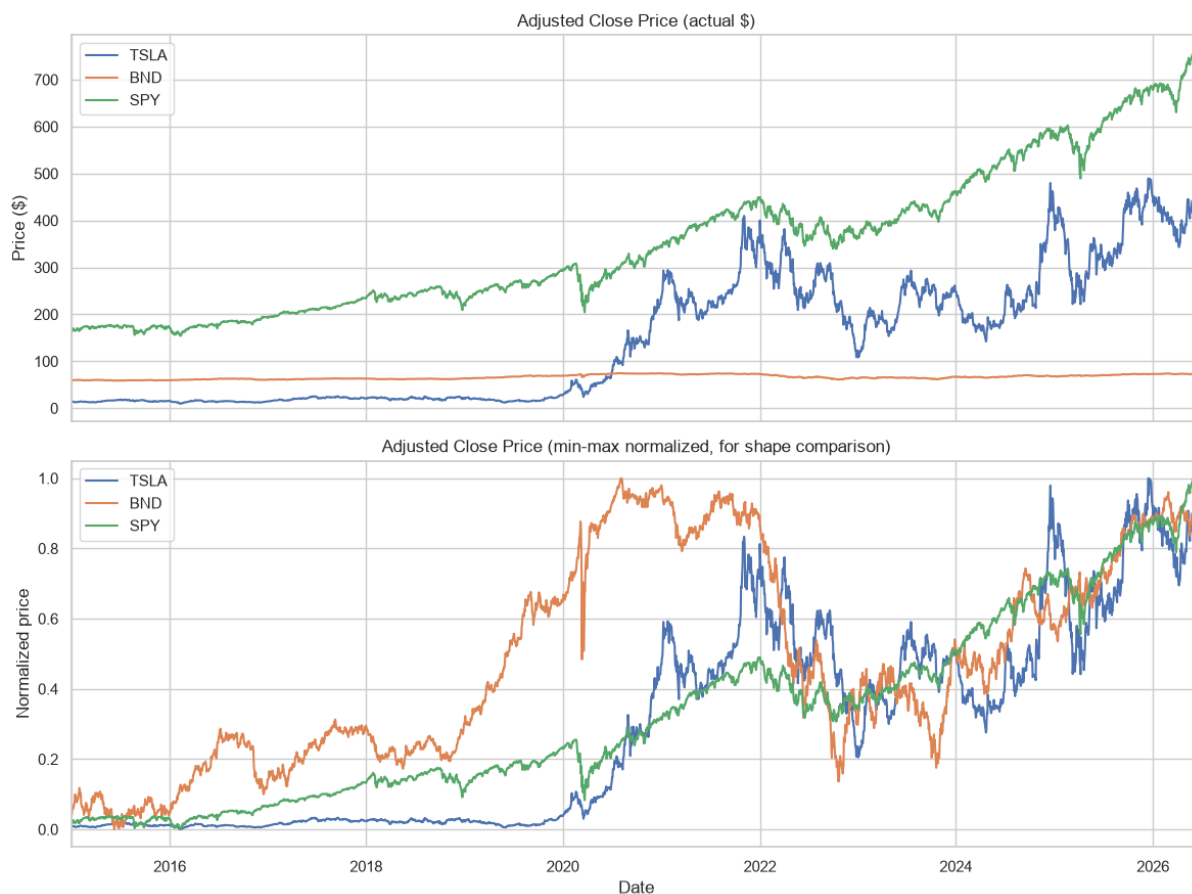


Figure 1. Adjusted close price — actual (top) and min-max normalized for shape comparison (bottom).

TSLA shows a strong overall upward trend from 2015 through 2026, punctuated by deep, multi-month drawdowns (notably 2022) — consistent with a high-growth/high-volatility profile. BND is essentially flat with low variance, reflecting bond price stability; SPY shows steady, moderate long-run appreciation.

2.2 Daily returns and volatility

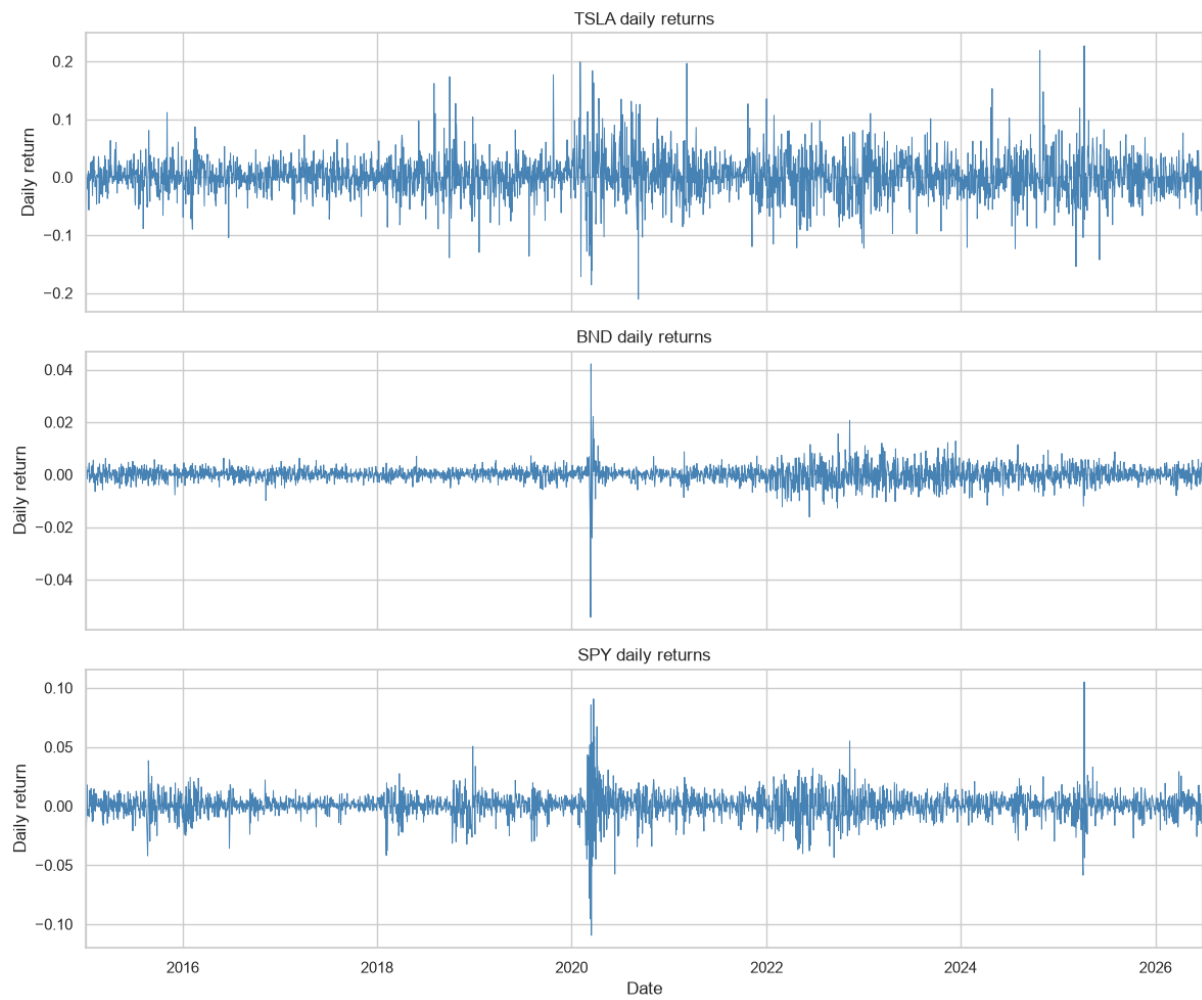


Figure 2. Daily percentage returns by asset.

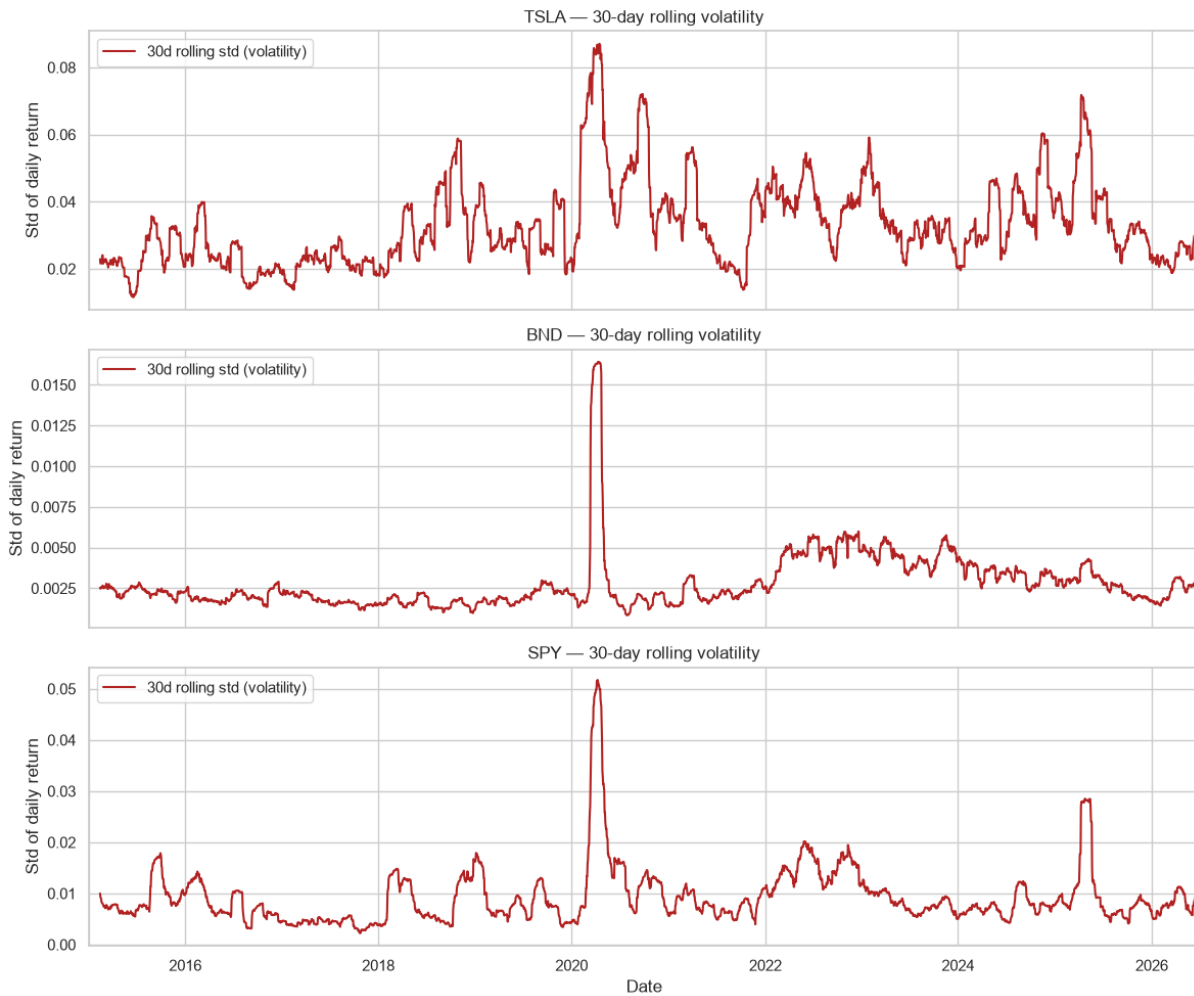


Figure 3. 30-day rolling volatility (standard deviation of daily returns).

TSLA's rolling volatility swings between calm and turbulent regimes rather than staying constant — large moves cluster together (e.g. around 2020 and 2022), a well-known feature of equity returns (volatility clustering) that a plain ARIMA model does not explicitly capture.

2.3 Outlier detection

Trading days with returns more than 3 standard deviations from each asset's mean:

Asset	Outlier days ($ \text{return} > 3\sigma$)	Most extreme single-day move
TSLA	50	+22.69% (2025-04-09)
BND	35	-5.44% (2020-03-12)
SPY	42	-10.94% (2020-03-16)

Extreme days cluster heavily around March 2020 (COVID-19 market shock) across all three assets, and around isolated earnings/macro events for TSLA specifically (e.g. 2025-04-09, 2024-10-24).

2.4 Correlation across assets

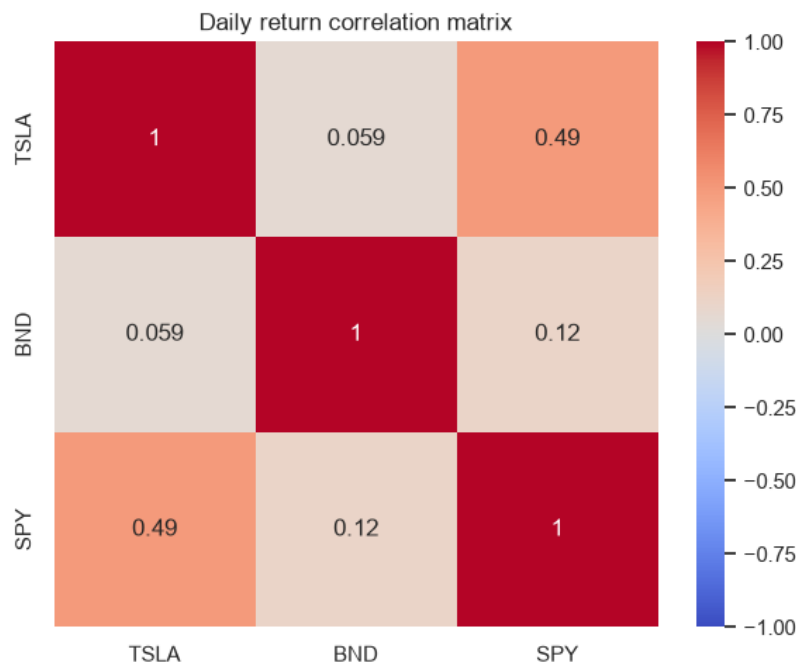


Figure 4. Daily return correlation matrix.

TSLA and SPY show moderate positive correlation (0.49); BND is nearly uncorrelated with both equities (0.059 with TSLA, 0.12 with SPY) — consistent with its intended role as a diversifying, stabilizing asset in the portfolio (relevant later for Task 4's Efficient Frontier).

3. Stationarity Testing (Augmented Dickey-Fuller)

ARIMA requires a stationary series. We test both the raw closing price level and the daily returns for each asset. **H0**: the series has a unit root (non-stationary). **H1**: the series is stationary. We reject H0 when p-value < 0.05.

Series	ADF statistic	p-value	Stationary at 5%?
TSLA — Close (level)	-1.04	0.739	No
TSLA — Daily return	-55.15	<0.001	Yes
BND — Close (level)	-1.12	0.708	No
BND — Daily return	-21.28	<0.001	Yes
SPY — Close (level)	1.43	0.997	No
SPY — Daily return	-15.70	<0.001	Yes

Interpretation: all three price *levels* fail to reject the ADF null — they are non-stationary, exhibiting a time-varying mean/trend consistent with a random walk, as the Efficient Market Hypothesis would predict. All three daily *return* series strongly reject the null and are stationary. This confirms a first difference (**d = 1**) is required before ARIMA can be validly applied to price levels — equivalently, returns can be modeled directly with **d = 0**.

4. Volatility and Risk Metrics

Annualized from daily returns (252 trading days/year), risk-free rate assumed 0%:

Asset	Annualized Return	Annualized Volatility	Historical VaR (95%, daily)	Sharpe Ratio
TSLA	43.8%	56.1%	5.11%	0.78
BND	1.9%	5.2%	0.47%	0.37
SPY	13.9%	17.3%	1.64%	0.80

TSLA carries by far the highest return *and* the highest risk (volatility and VaR) of the three assets — a 95% historical VaR of 5.11% means TSLA's daily loss is not expected to exceed 5.11% on 95% of trading days, roughly 10× BND's threshold. BND anchors the portfolio with minimal downside risk but a correspondingly small return contribution. SPY sits between the two on every metric, as expected for a broad, diversified index.

5. Progress on Task 2 — ARIMA Baseline

The TSLA adjusted-close series was split **chronologically** — no shuffling, preserving temporal order throughout: **train** = 2015-01-02 to 2024-12-31 (2,608 observations), **test** = 2025-01-01 to 2026-06-29 (391 observations).



Figure 5. Chronological train/test split for TSLA adjusted close.

`pmdarima.auto_arima` (stepwise search, non-seasonal, `max_p=5`, `max_q=5`, `max_d=2`) was used to select the order. The search converged on **ARIMA(0,1,0) with no drift** — a driftless random walk on the price level. This is a meaningful result on its own: the best-fitting classical linear model finds no exploitable autocorrelation structure in TSLA's past prices beyond the previous day's value, exactly as the Efficient Market Hypothesis predicts.

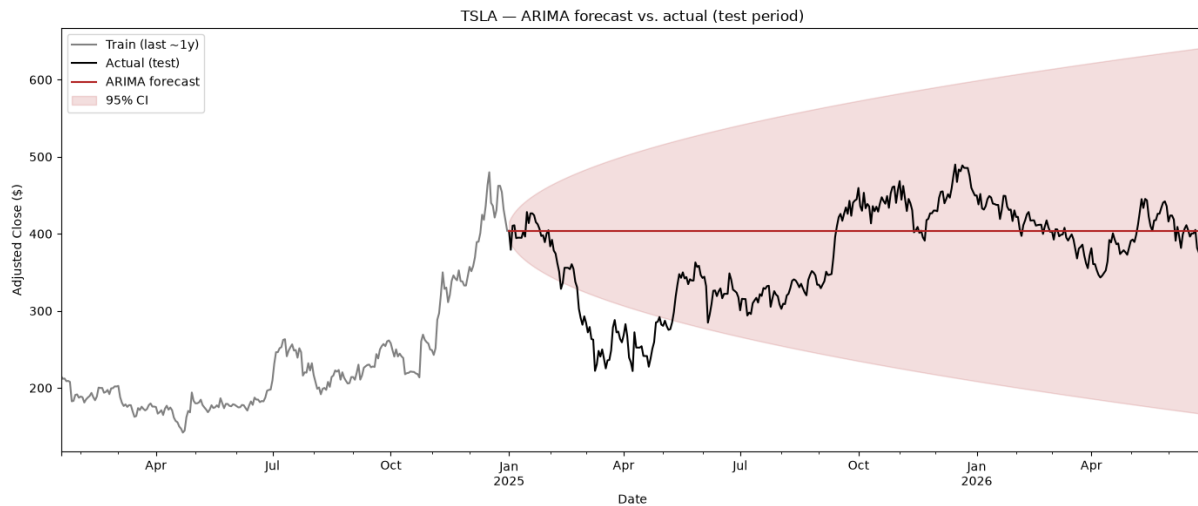


Figure 6. ARIMA(0,1,0) forecast vs. actual TSLA price over the test period, with 95% confidence interval.

Model	MAE	RMSE	MAPE
ARIMA(0,1,0)	54.15	70.20	17.11%

Reading the forecast chart: the flat red forecast line and the steadily widening confidence band are direct visual evidence that a static, driftless random-walk model has no mechanism for predicting direction beyond the last training price — all of the growth/decline TSLA actually experienced during the 18-month test window shows up as forecast error. This baseline is the reference point against which the LSTM model (planned next) will be compared.

6. Next Steps (Toward Final Submission)

- Build and tune an LSTM model on the same chronological split for direct comparison with the ARIMA baseline (Task 2).
- Generate 6–12 month forward forecasts with confidence intervals from the best-performing model (Task 3).
- Feed the TSLA forecast into a Modern Portfolio Theory optimization alongside BND/SPY historical returns, and produce the Efficient Frontier (Task 4).
- Backtest the resulting optimal portfolio against a static 60/40 SPY/BND benchmark (Task 5).